

DATA SCIENCE RESOURCES

Unsupervised Learning Projects

Project 1: Customer Segmentation Using K-Means

Problem Statement: Businesses struggle to personalize marketing and product strategies for diverse customer bases. A one-size-fits-all approach reduces engagement and retention.

Objective: Group customers based on purchasing behavior, demographics, and activity using K-Means Clustering to enable targeted marketing strategies.

Dataset Source

- [Mall Customer Segmentation Dataset \(Kaggle\)](#)
- [Retail CRM export files or e-commerce user logs](#)

Techniques and Concepts

- Data normalization (MinMaxScaler, StandardScaler)
- Elbow method and silhouette score for optimal k
- K-Means clustering
- PCA for visualization of clusters

Tools and Libraries: Python, pandas, scikit-learn, matplotlib, seaborn, plotly

Expected Outcome: Visual cluster groups that help businesses identify high-value customers, discount seekers, or infrequent shoppers for campaign targeting.

Project 2 : Movie Genre Clustering

Problem Statement: Streaming platforms need to understand movie similarity and genre overlap to improve recommendations and navigation.

Objective: Cluster movies based on metadata (tags, synopsis, cast, ratings) to find similar titles or hidden genre combinations.

Dataset Source

- [TMDB 5000 Movie Dataset](#)
- [IMDb + MovieLens Merge](#)

Techniques and Concepts

- Text vectorization (TF-IDF on synopsis)
- Feature encoding (genres, ratings, runtime)
- Dimensionality reduction (PCA, t-SNE)
- K-Means, Hierarchical Clustering

Tools and Libraries: Python, scikit-learn, pandas, NLTK, spaCy, plotly

Expected Outcome: A genre clustering system that shows which movies are similar based on storyline, cast, and themes — useful for content-based filtering.

Project 3 : Market Basket Analysis

Problem Statement: Retailers want to understand which items are frequently purchased together to optimize product placement, bundling, and promotions.

Objective: Identify item associations and clusters in transactional data using association rules and clustering.

Dataset Source: [Instacart Market Basket Data](#)

Techniques and Concepts

- Apriori algorithm for association rules
- Itemset frequency filtering
- K-Modes or DBSCAN for categorical clustering
- Lift, confidence, and support metrics

Tools and Libraries: Python, mlxtend, pandas, seaborn, matplotlib

Expected Outcome: Association rules like “If a customer buys milk, they are likely to buy bread,” and product clusters for bundle offers.

Project 4 : Crime Rate Clustering in Indian States

Problem Statement: Law enforcement and policy-makers need to analyze regional crime trends to allocate resources, enhance security, and monitor high-risk zones.

Objective: Cluster Indian states based on crime rates across various categories like theft, assault, and cybercrime using unsupervised learning.

Dataset Source

- [National Crime Records Bureau \(NCRB\)](#)
- [India Crime Statistics](#)

Techniques and Concepts

- Feature normalization
- K-Means or Agglomerative Clustering
- Heatmaps and cluster maps for visualization
- Interpretation of regional crime patterns

Tools and Libraries: Python, pandas, seaborn, scikit-learn, geopandas (for maps)

Expected Outcome: Interactive visualizations and clusters showing which Indian states have similar crime profiles for decision-making and policing.
